

Problem

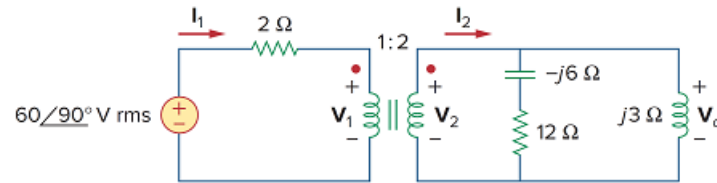
For the ideal transformer circuit of Fig. 13.122 below, find:

(a) I_1 and I_2 ,

(b) V_1 , V_2 , and V_o ,

(c) the complex power supplied by the source.

Figure 13.122



Step-by-step solution

Step 1 of 3

$$\begin{aligned} \text{(A) } Z_L &= j3 \parallel (12 - j6) = \frac{j3(12 - j6)}{(12 - j3)} \\ &= \frac{(12 + j54)}{17} \end{aligned}$$

Reflecting this to the primary side gives,

$$\begin{aligned} Z_{in} &= 2 + \frac{Z_L}{n^2} = 2 + \frac{(3 + j13.5)}{17} \\ &= 2.3168 \angle 20.04^\circ \\ I_1 &= \frac{V_s}{Z_{in}} = \frac{60 \angle 90^\circ}{2.3168 \angle 20.04^\circ} \\ &= 25.9 \angle 69.96^\circ \text{ A (rms)} \\ I_2 &= \frac{I_1}{n} = 12.95 \angle 69.96^\circ \text{ A (rms)} . \end{aligned}$$

Step 2 of 3

$$\begin{aligned} \text{(B) } 60 \angle 90^\circ &= 2I_1 + V_1 \quad \text{Or} \quad V_1 = V_1 = j60 - 2I_1 \\ &= j60 - 51.8 \angle 69.96^\circ \\ V_1 &= 21.06 \angle 147.44^\circ \text{ V (rms)} \\ V_2 &= nV_1 = 42.12 \angle 147.44^\circ \text{ V (rms)} \\ V_o &= V_2 = 42.12 \angle 147.44^\circ \text{ V (rms)} . \end{aligned}$$

Step 3 of 3

$$\begin{aligned} \text{(C) } S &= V_s I_1^* = (60 \angle 90^\circ) (25.9 \angle -69.96^\circ) \\ &= 1554 \angle 20.04^\circ \text{ V} . \end{aligned}$$

[Comments \(1\)](#)



Anonymous

the unit for complex power should be (VA)
